

1. Supply Chain Tracking System

Abstract

The Supply Chain Tracking System is a decentralized application (DApp) developed using blockchain technology to monitor and manage the movement of products across the supply chain. The system uses Ethereum blockchain, Solidity smart contracts, and MetaMask wallet integration to ensure transparency, security, and data integrity.

In traditional supply chain systems, product information is stored in centralized databases, which can be modified or manipulated by unauthorized users. This creates issues such as lack of transparency, fake products, and difficulty in tracking goods. The proposed blockchain-based system solves these problems by storing product transaction records on a decentralized blockchain network where data cannot be altered once recorded.

The application allows manufacturers, suppliers, distributors, and retailers to update product status and track product movement in real time. Each transaction is securely verified and permanently stored on the blockchain, ensuring trust among all participants. The system improves product traceability, reduces fraud, and enhances overall supply chain efficiency.

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2. Blockchain Voting for College Elections

Abstract

The Blockchain Voting for College Elections is a decentralized voting application developed using Ethereum blockchain technology to provide a secure, transparent, and tamper-proof election system for colleges. The project uses Solidity smart contracts, MetaMask wallet authentication, and a web-based frontend interface for conducting digital elections.

Traditional voting systems may suffer from issues such as vote manipulation, fake voting, lack of transparency, and centralized control. The proposed

blockchain-based voting system addresses these challenges by storing voting records on a decentralized blockchain network where votes cannot be modified or deleted after submission.

In this system, students can securely log in using their blockchain wallets and cast votes for candidates. Each student is allowed to vote only once, ensuring fairness in the election process. The system automatically counts votes in real time and displays transparent election results.

By using blockchain technology, the project enhances election security, transparency, and trust among students and administrators.

3. Decentralized Donation System

Abstract

The Decentralized Donation System is a blockchain-based application designed to provide a secure and transparent platform for online donations. The system uses Ethereum blockchain technology, Solidity smart contracts, and MetaMask wallet integration to manage donation transactions without relying on centralized authorities.

In traditional donation platforms, users may face issues related to lack of transparency, misuse of funds, and hidden transaction details. The proposed decentralized system solves these problems by recording all donation transactions on the blockchain, allowing anyone to verify donation history publicly.

The application enables users to donate cryptocurrency directly to charity organizations through smart contracts. All donation records are securely stored on the blockchain, ensuring transparency and preventing unauthorized modifications. The system also allows tracking of total donations and fund usage in real time.

By using blockchain technology, the project increases trust, security, and transparency in charity and fundraising activities.